



Motivation

- Carbapenem resistant *Acinetobacter baumannii* (CRAB) is a bacterium of global health concern, and is a particular threat to vulnerable patients in intensive care units (ICUs).
- Bacteria can spread from the hospital environment to patients, which needs to be understood and controlled to reduce hospital-acquired infections.
- We model a longitudinal data set of diagnostic tests for both patients and bed units.
- We can model the spread of bacteria using a coupled hidden Markov model, and we fit this in a Bayesian framework using the forward filtering backward sampling (iFFBS) algorithm.

Coupled Hidden Markov Model

We consider patients and bed units as two different “types” of individuals. We consider colonisation between individuals in the same room, with colonisation rates shown below.

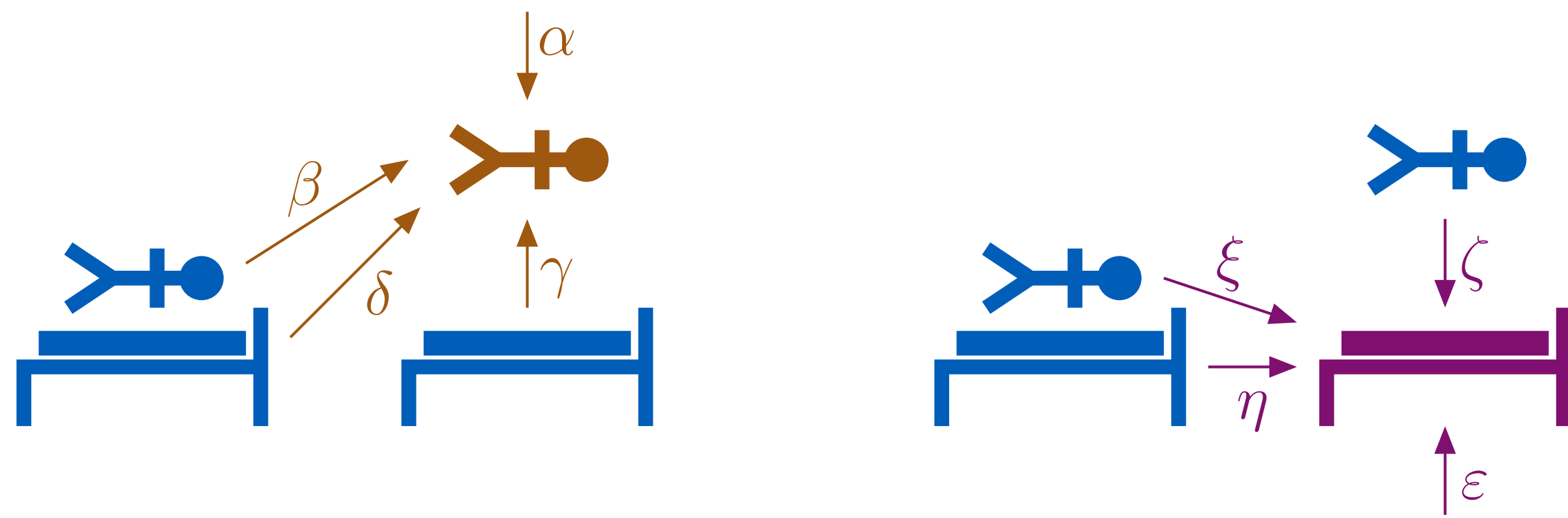


Figure 1. Left: A patient in the ICU is colonised by: external sources at rate α ; other patients at rate β ; the bed unit occupied by the patient at rate γ ; and other bed units at rate δ . Right: A bed unit in the ICU is colonised by: external sources at rate ϵ ; the patient occupying it at rate ζ ; patients in other bed units at rate ξ ; and other bed units at rate η .

We use the colonisation rates to derive the probability of patient p avoiding colonisation from day t to day $t + 1$ as

$$q_t^{(p)} = \exp \left(-\alpha - \underbrace{\beta P_t^{(-p)}}_{\text{other patients}} - \underbrace{\gamma B_t^{(p)}}_{\text{patient's bed unit}} - \underbrace{\delta B_t^{(-p)}}_{\text{other bed units}} \right).$$

Similarly, the probability of bed unit b avoiding colonisation from day t to day $t + 1$ is

$$q_t^{(b)} = \exp \left(-\epsilon - \underbrace{\xi P_t^{(-b)}}_{\text{other patients}} - \underbrace{\zeta P_t^{(b)}}_{\text{bed unit's patient}} - \underbrace{\eta B_t^{(-b)}}_{\text{other bed units}} \right).$$

On day t , we observe a diagnostic test result $Y_t^{(p)}$ or $Z_t^{(b)}$ which is dependent on the individual's true hidden state $P_t^{(p)}$ or $B_t^{(b)}$. We can model the diagnostic tests using a hidden Markov model:

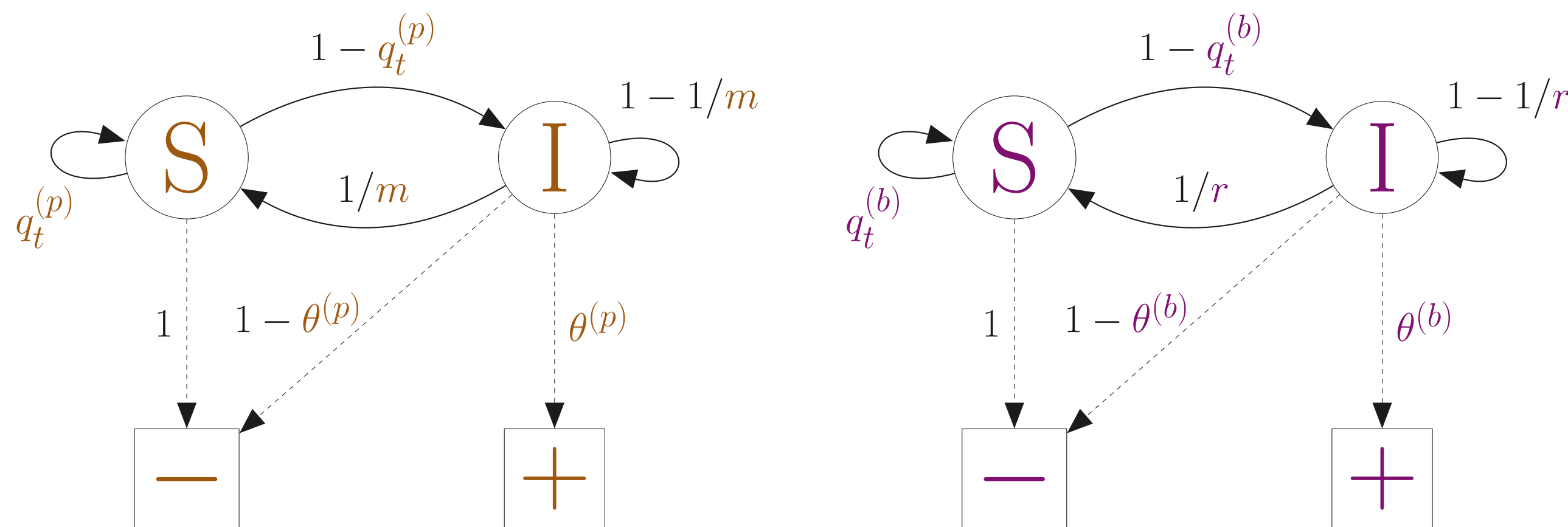
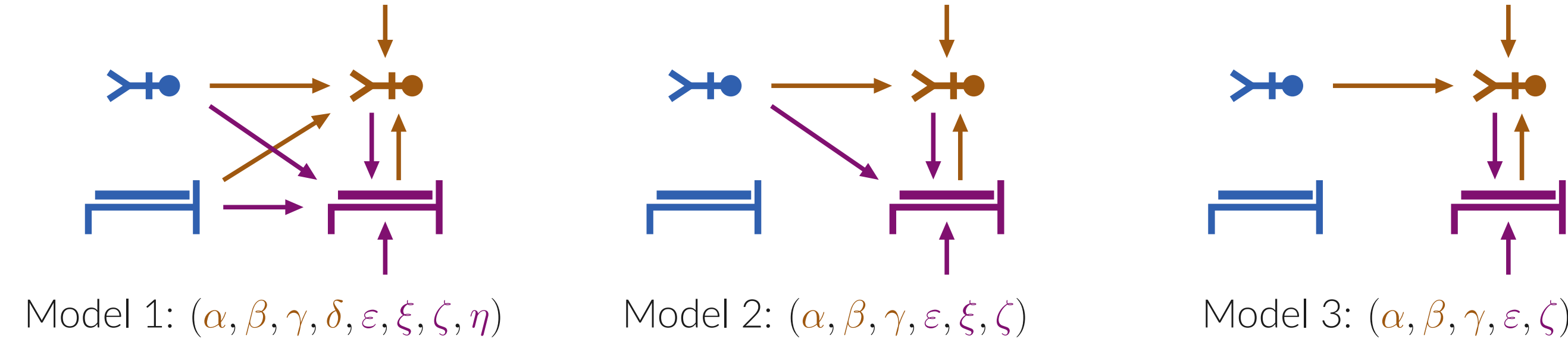


Figure 2. Left: Hidden Markov Model for a single patient in the ICU. Right: Hidden Markov Model for a single bed unit. Note that the transition probabilities $S \rightarrow S$ and $S \rightarrow I$ are dependent on the states of the other chains.

This setup means the mean colonisation period is m days for patients and r days for bed units, and the diagnostic test sensitivities are $\theta^{(p)}$ and $\theta^{(b)}$. Patients are colonised on entry to the ICU with probability $\nu^{(p)}$ and bed units are colonised on the first day with probability $\nu^{(b)}$.

Models of Bacteria Spread

We consider three different models with different colonisation rates, shown in the diagram below. All three models contain the other parameters alongside the specified colonisation rates.



Experimental Data Results

Estimated Median and 95% C.I. for Each Parameter

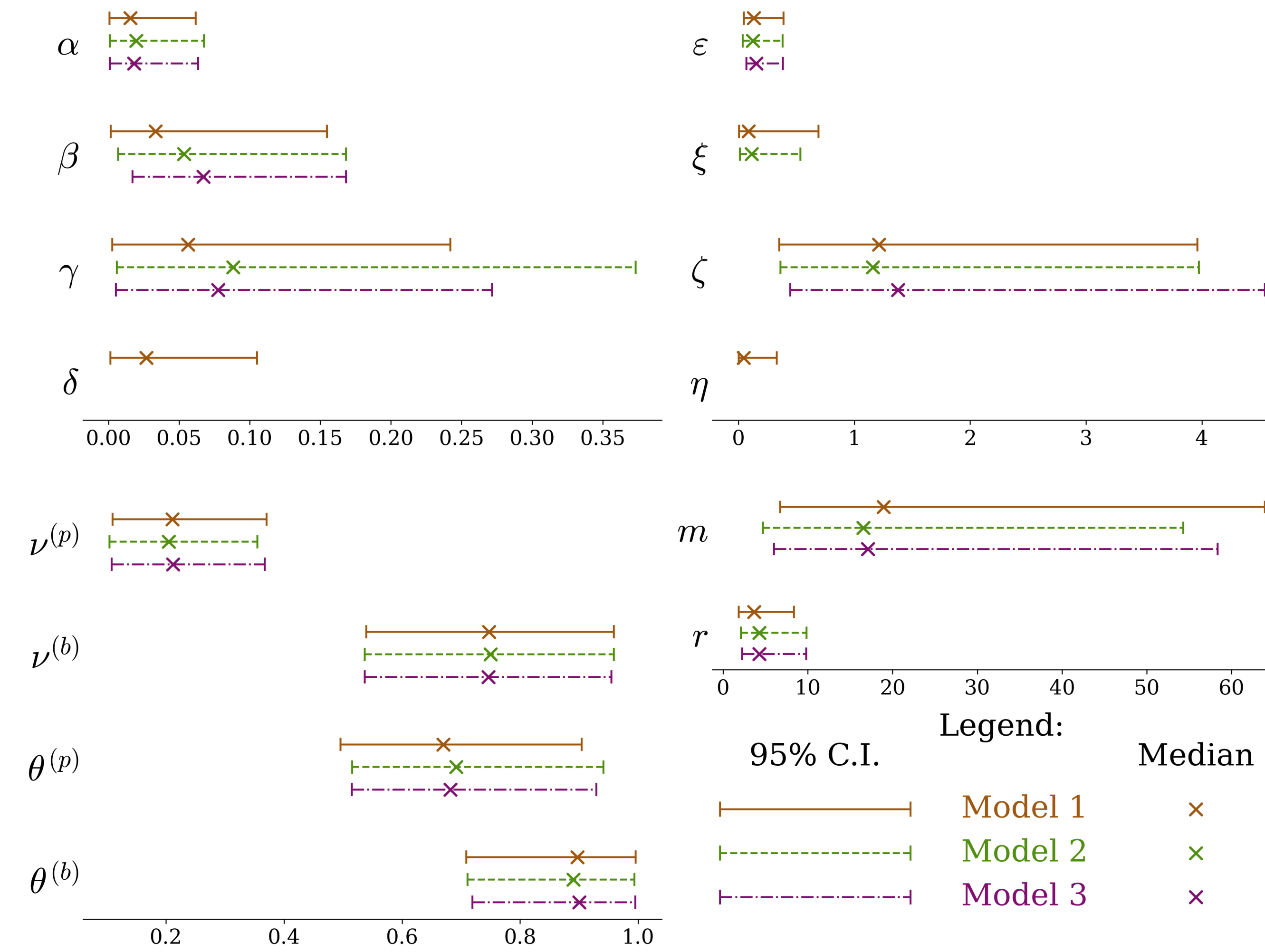


Figure 3. Median and 95% credible interval for each parameter under the three models.

- Overlapping credible intervals suggest a lack of significant differences between patient colonisation rates, but β and γ are higher than α and δ , showing patients are colonised by bacteria from their immediate environment.
- Estimates for parameters other than colonisation rates are robust to the model specification, validating our model.

Parameter Inference

Given the observed data, we augment the model with $\mathbf{P}_{1:T}^{1:N_p}$ and $\mathbf{B}_{1:T}^{1:N_b}$ as additional parameters to be estimated. We use MCMC and the iFFBS algorithm to sample from the posterior distribution for $\mathbf{P}_{1:T}^{1:N_p}$, $\mathbf{B}_{1:T}^{1:N_b}$, and the model parameters φ . The iFFBS algorithm allows us to sample

$$\mathbf{P}_{1:T}^{(p)} \sim \pi \left(\mathbf{P}_{1:T}^{(p)} | \mathbf{X}_{1:T}^{(p)}, \mathbf{P}_{1:T}^{(-p)}, \mathbf{B}_{1:T}^{1:N_b}, \varphi \right) \quad \text{and} \quad \mathbf{B}_{1:T}^{(b)} \sim \pi \left(\mathbf{B}_{1:T}^{(b)} | \mathbf{Y}_{1:T}^{(b)}, \mathbf{B}_{1:T}^{(-b)}, \mathbf{P}_{1:T}^{1:N_p}, \varphi \right)$$

to update the hidden states for patient p and bed unit b . We repeat this for $p = 1, \dots, N_p$ and $b = 1, \dots, N_b$, which generates a new sample of $\mathbf{P}_{1:T}^{1:N_p}$ and $\mathbf{B}_{1:T}^{1:N_b}$.

We update parameters $(\alpha, \beta, \gamma, \delta)$ and $(\epsilon, \xi, \zeta, \eta)$ with two separate Hamiltonian steps.

We update parameters $(m, r, \nu^{(p)}, \nu^{(b)}, \theta^{(p)}, \theta^{(b)})$ with individual Gibbs steps.

Simulation Study

We ran a simulation study informed by the real dataset which we are using to develop this model.

We estimate patient parameters well, and struggle with bed unit parameters. Our model is still useful from a clinical standpoint to understand how patients can be colonised by CRAB.

Boxplots of Estimated Parameter Medians

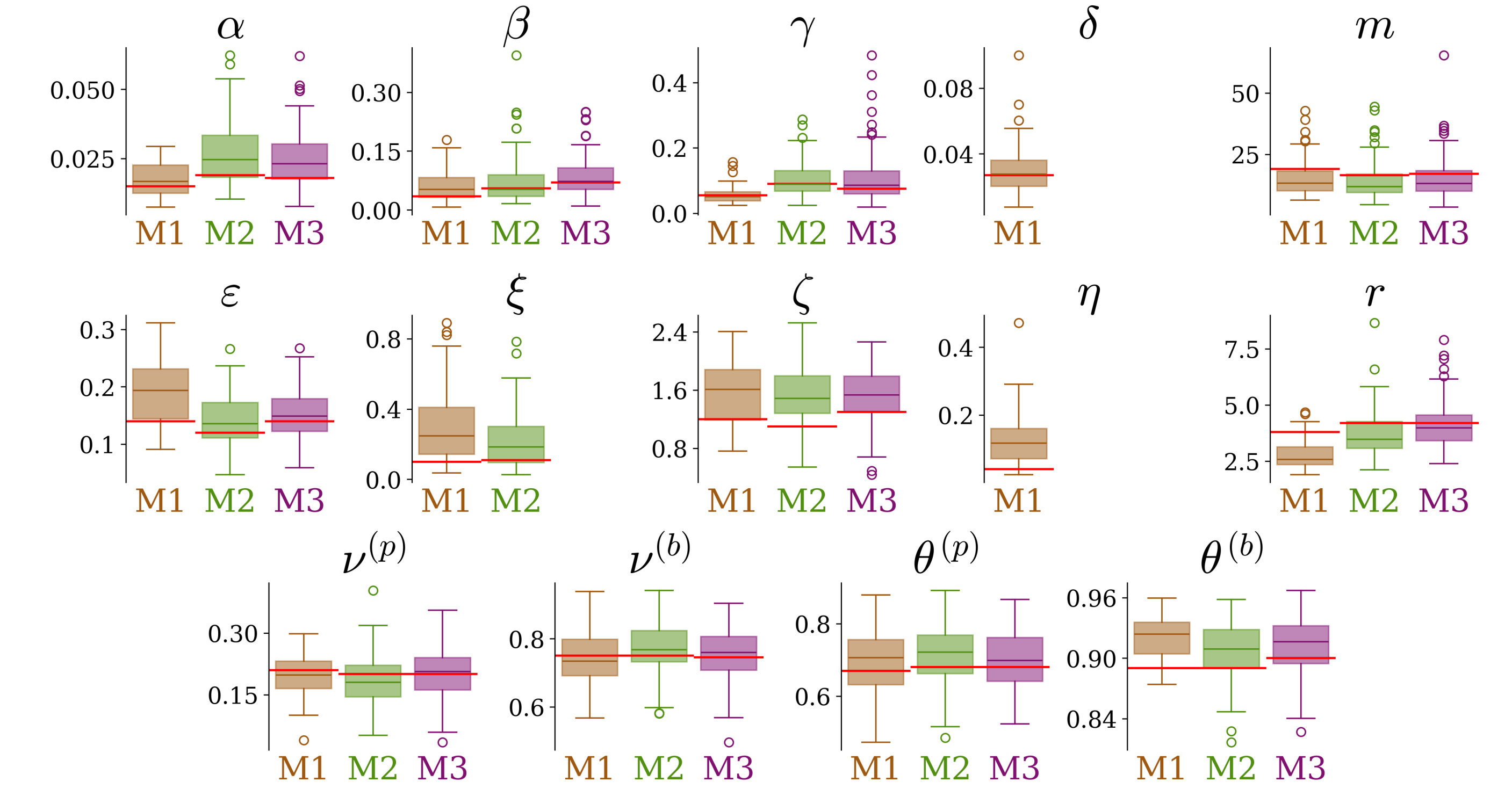


Figure 4. Boxplots of estimated posterior medians, with the true value of the parameter shown as a horizontal line.

Model Limitations

- MCMC performance is impacted by the prevalence, sensitivity, and test interval.
- We can use the effective sample size (ESS) to measure the performance of the MCMC algorithm for datasets with varying prevalence and sensitivity.
- This can inform future data collection to ensure our methods can be applied, by suggesting more frequent testing or a more sensitive test for a given prevalence level.

ESS of Estimated Prevalence

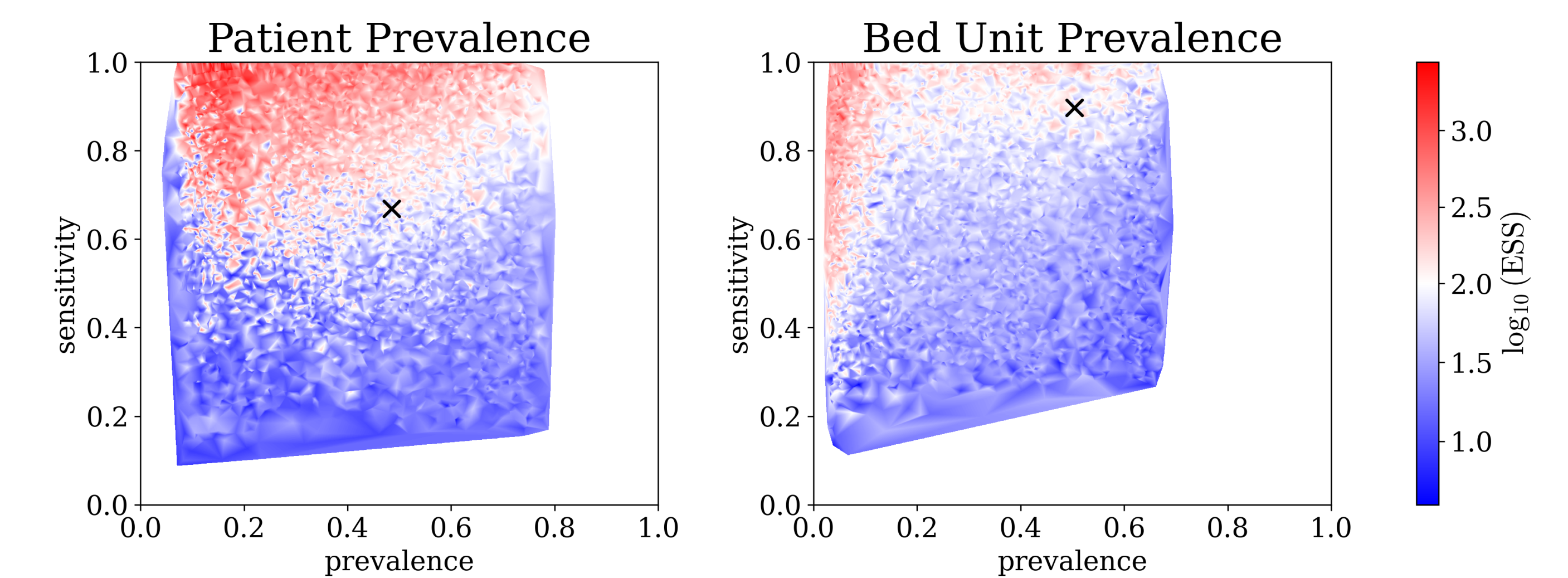


Figure 5. Heatmaps of the ESS of estimated prevalence for datasets with different prevalence and sensitivity. Red areas are where inference performs best, and crosses mark the estimated position of our experimental dataset.

Conclusions and Future Work

- Our findings have clinical implications for controlling CRAB and other bacteria.
- Our methods can be tailored to specific hospital layouts to suggest effective interventions.
- We can understand the model limitations, and use this to inform future data collection.
- Future work will use model selection methods to select the colonisation rates.